

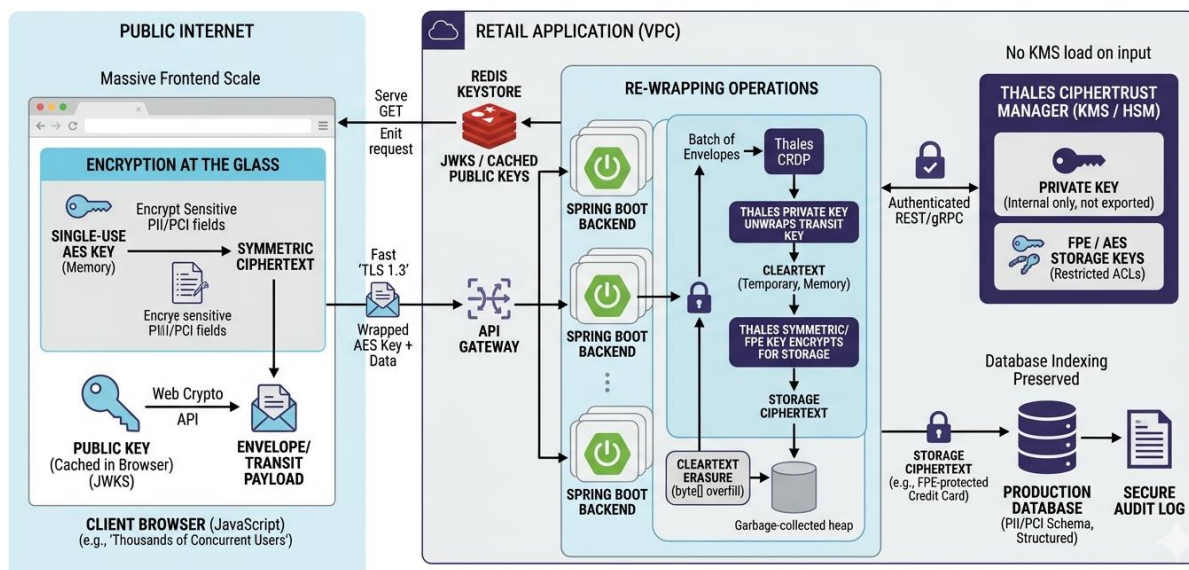
Thales Client Side End to End Protection.

This document describes how Thales CipherTrust Manager can be used to protect sensitive data exactly *where* it is captured: on the client device (browser, mobile app, IoT device) rather than the server.

Another term commonly used to describe this capability is **"At the glass"** means **securing sensitive data right where the user's fingertips touch the screen**. It refers to executing encryption directly inside the client's web browser or mobile app interface, *before* the data is even committed to the local device's memory stack or transmitted across a network interface.

Transit-to-Storage Re-wrapping Pattern with Thales CipherTrust

Comprehensive Data Protection Pipe- Retail Web Application



The above diagram explains the end-to-end data lifecycle, highlighting how to use Thales CipherTrust to secure data both in transit and at rest, while maintaining high performance.

How it works

1. Security at the Source (The Browser)

Protect the data **"at the glass"** the instant the user enters it.

- The browser automatically creates its own single-use symmetric key.
- It encrypts the PII/PCI immediately. It doesn't use the network yet.
- It then "wraps" (envelopes) that single-use key with our highly scalable public key.

- **Business Value:** If the network connection is intercepted, or if malicious software is sitting between the user and our backend, the data is useless. The "cleartext" never leaves the user's device.

2. Scaling the Front Door (The Redis Caching Layer)

The **Redis Keystore** section, which addresses the performance concerns.

- Rather than making the browser talk to the Thales KMS every single time a page loads, our backend pre-fetches the *public keys* and stores them in ultra-fast Redis memory.
- The backend serves thousands of public key requests to users instantly (milliseconds).
- **Business Value:** We can scale to thousands of simultaneous retail users without slowing down the website or putting any "crypto load" on the security infrastructure.

3. The Trusted Re-wrap (The Backend Transition)

When the wrapped payload hits the backend:

- The **Spring Boot** application identifies the payload and securely authenticates with the **Thales CipherTrust Manager** using private, backend-only credentials.
- The backend will "unwrap" the temporary key and immediately re-encrypt the data with our permanent, structured storage key (preferably using **Format-Preserving Encryption - FPE**).
- **The "Micro-exposure" window:** The data is only cleartext for a few *milliseconds* in server memory during this re-wrap, and then it is immediately erased.
- **Business Value:** The Spring Boot app never writes cleartext to a log file or temp database. The database receives a perfectly formatted, encrypted value (e.g., a 16-digit fake number instead of a real one), allowing us to preserve database indexing and schema performance without compromising security.

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Summary

This design gives you security postures at the frontend using your standard Thales backend infrastructure, solved with modern Web Crypto APIs and intelligent caching.